Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 175
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An application that is deployed to Amazon EC2 is using Amazon DynamoDB. The application calls the DynamoDB REST API. Periodically, the application receives a ProvisionedThroughputExceededException error when the application writes to a DynamoDB table.

Which solutions will mitigate this error MOST cost-effectively? (Choose two.)

A. Modify the application code to perform exponential backoff when the error is received.

B. Modify the application to use the AWS SDKs for DynamoDB.

C. Increase the read and write throughput of the DynamoDB table.

D. Create a DynamoDB Accelerator (DAX) cluster for the DynamoDB table.

E. Create a second DynamoDB table. Distribute the reads and writes between the two tables.

Show Suggested Answer

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by fordiscussionstwo at *Oct. 6, 2023, 2:28 a.m.*

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Comments

SerialiDr Highly Voted 2 years, 2 months ago

Selected Answer: AB

A. Modify the application code to perform exponential backoff when the error is received: This is a cost-effective and recommended approach. Exponential backoff is a standard error-retry strategy where the time between retries gradually increases. This strategy helps to efficiently manage request retries without immediately consuming additional throughput, thus reducing the likelihood of repeatedly hitting the throughput limits.

B. Modify the application to use the AWS SDKs for DynamoDB: The AWS SDKs implement best practices, including automatic retry logic with exponential backoff. Using an AWS SDK for DynamoDB can simplify the implementation and is more efficient than directly calling the DynamoDB REST API. This change can help mitigate throughput exceedance errors.

upvoted 6 times

f271c23 Most Recent 1 year, 3 months ago

Selected Answer: AB

Yes AB is the right set of answers.

upvoted 1 times

CloudChingon 1 year, 4 months ago

Key Words Why They Matter
ProvisionedThroughputExceededException Indicates the application is exceeding the provisioned capacity of the DynamoDB table.

Periodically Suggests that the issue happens under specific conditions, such as traffic spikes.

REST API Implies the application is making direct API calls, possibly without SDK optimizations.

Cost-effectively Points towards solutions that avoid increasing DynamoDB capacity (which costs more).
Writes to a DynamoDB table
Identifies the operation likely causing the throttling issue.

Modify application code Indicates that a code change is necessary to handle throttling errors better.

Exponential backoff A retry strategy to reduce the impact of throttling.

AWS SDKs Built-in solutions for efficient, optimized interactions with DynamoDB.

upvoted 1 times

aws_god 1 year, 5 months ago

Selected Answer: BC

This is what the exception looks like:
"boto.dynamodb2.exceptions.ProvisionedThroughputExceededExceptio
n: ProvisionedThroughputExceededException: 400 Bad Request
{u'message': u'The level of configured provisioned throughput for the table was exceeded. Consider increasing your provisioning level with the

UpdateTable API', u'__type':
u'com.amazonaws.dynamodb.v20120810#ProvisionedThroughputExceededException'}"

You need B since the AWS SDK includes automatic retry logic with exponential backoff specifically designed to deal with this automatically and C since increasing the provisioned level also helps solve this issue.

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: AB

AB is the correct answer.

upvoted 1 times

TanTran04 2 years, 3 months ago

Selected Answer: AC

Following ChatGPT 3.5, Option A and C
Option B (Modify the application to use the AWS SDKs for DynamoDB) is not directly related to resolving throughput issues. It's generally recommended to use the AWS SDKs as they provide more efficient and convenient ways to interact with AWS services, but it may not directly address the ProvisionedThroughputExceededException issue.

upvoted 1 times

dilleman 2 years, 5 months ago

Selected Answer: AB

A and B.
Exponential backoff is a standard error-handling strategy for network applications. The idea is to retry a failed request with increasing delays between each attempt.

And the AWS SDKs have built-in support for handling these errors.

upvoted 4 times

Digo30sp 2 years, 5 months ago

Selected Answer: AB

A and B: https://www.examtopics.com/discussions/amazon/view/69199-exam-aws-certified-developer-associate-topic-1-question-385/

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